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Advanced Control Systems

Module III

Part I

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Syllabus

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Controllability & Observability

- A system is said to be **controllable** at time t_0 if it is possible by means of an unconstrained control vector to transfer the system from any initial state $\mathbf{x}(t_0)$ to any other state in a finite interval of time.
- A system is said to be **observable** at time t_0 if, with the system in state $\mathbf{x}(t_0)$, it is possible to determine this state from the observation of the output over a finite time interval.

Kalman's Test to check controllability

The result just obtained can be extended to the case where the control vector $\mathbf{u}$ is r -dimensional. If the system is described by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

where $\mathbf{u}$ is an r -vector, then it can be proved that the condition for complete state controllability is that the $n \times nr$ matrix

$$[\mathbf{B} \mid \mathbf{AB} \mid \cdots \mid \mathbf{A}^{n-1}\mathbf{B}]$$

be of rank n , or contain n linearly independent column vectors. The matrix

$$[\mathbf{B} \mid \mathbf{AB} \mid \cdots \mid \mathbf{A}^{n-1}\mathbf{B}]$$

is commonly called the *controllability matrix*.

Kalman's Test to check controllability

$$Q_c = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} \quad \dots (5.64)$$

where n is the order of the system (n is also equal to number of state variables)

In this case the system is completely state controllable if the rank of the composite matrix, Q_c is n .

The rank of the matrix is n , if the determinant of ($n \times n$) composite matrix Q_c is non-zero. i.e, if $|Q_c| \neq 0$, then rank of $Q_c = n$ and the system is completely state controllable.

Examples for controllability test

Consider the system given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

Since

$$[\mathbf{B} \mid \mathbf{AB}] = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \text{singular}$$

the system is not completely state controllable.

Consider the system given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

For this case,

$$[\mathbf{B} \mid \mathbf{AB}] = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} = \text{nonsingular}$$

The system is therefore completely state controllable.

Kalman's test for observability

If the system is completely observable, then, given the output $\mathbf{y}(t)$ over a time interval $0 \leq t \leq t_1$, $\mathbf{x}(0)$ is uniquely determined from Equation (9-65). It can be shown that this requires the rank of the $nm \times n$ matrix

$$\begin{bmatrix} \mathbf{C} \\ \cdots \\ \mathbf{CA} \\ \cdots \\ \mathbf{CA}^{n-1} \end{bmatrix}$$

to be n . (See Problem **A-9-19** for the derivation of this condition.)

From this analysis, we can state the condition for complete observability as follows: The system described by Equations (9-63) and (9-64) is completely observable if and only if the $n \times nm$ matrix

$$[\mathbf{C}^* \mid \mathbf{A}^* \mathbf{C}^* \mid \cdots \mid (\mathbf{A}^*)^{n-1} \mathbf{C}^*]$$

is of rank n or has n linearly independent column vectors. This matrix is called the *observability matrix*.

Problem to check observability

Show that the following system is not completely observable:

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$y = \mathbf{Cx}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [4 \quad 5 \quad 1]$$

Problem to check observability

Note that the control function u does not affect the complete observability of the system. To examine complete observability, we may simply set $u = 0$. For this system, we have

$$[\mathbf{C}^* \mid \mathbf{A}^*\mathbf{C}^* \mid (\mathbf{A}^*)^2\mathbf{C}^*] = \begin{bmatrix} 4 & -6 & 6 \\ 5 & -7 & 5 \\ 1 & -1 & -1 \end{bmatrix}$$

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Note that

$$\begin{vmatrix} 4 & -6 & 6 \\ 5 & -7 & 5 \\ 1 & -1 & -1 \end{vmatrix} = 0$$

Hence, the rank of the matrix $[\mathbf{C}^* \mid \mathbf{A}^*\mathbf{C}^* \mid (\mathbf{A}^*)^2\mathbf{C}^*]$ is less than 3. Therefore, the system is not completely observable.

Problem

EXAMPLE 5.6

Write the state equations for the system shown in fig 5.6.1 in which x_1 , x_2 and x_3 constitute the state vector. Determine whether the system is completely controllable and observable.

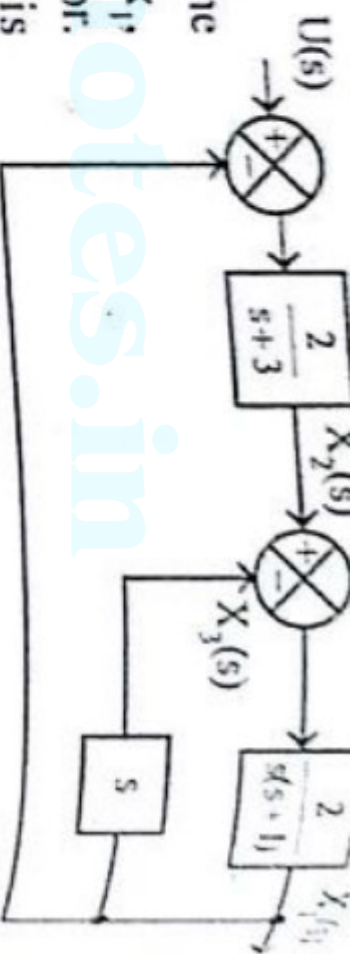


Fig. 5.6.1

Problem

SOLUTION

To find state model

The state equations are obtained by writing equations for the output of each block and then taking inverse Laplace transform.

With reference to fig 5.6.2 we can write,

$$X_1(s) = [X_2(s) - X_3(s)] \left[\frac{2}{s(s+1)} \right]$$

$$s(s+1) X_1(s) = 2X_2(s) - 2X_3(s)$$

$$s^2 X_1(s) + s X_1(s) = 2X_2(s) - 2X_3(s)$$

On taking inverse laplace transform,

$$\ddot{x}_1 + \dot{x}_1 = 2x_2 - 2x_3 \quad \dots\dots(5.6.1)$$

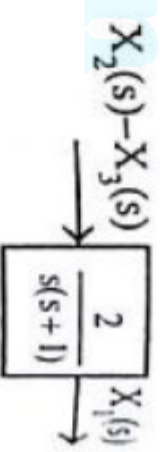
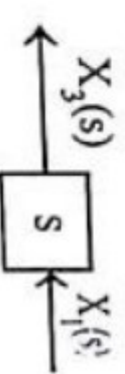


Fig 5.6.2



Problem

On taking inverse laplace transform,

$$\ddot{x}_1 + \dot{x}_1 = 2x_2 - 2x_3 \quad \text{.....(5.6.1)}$$

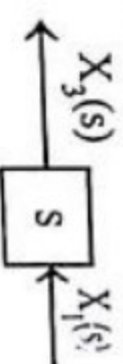


Fig 5.6.3

With reference to fig 5.6.3, we can write,

$$X_3(s) = sX_1(s)$$

On taking inverse laplace transform

$$x_3 = \dot{x}_1 \quad \text{.....(5.6.2)}$$

With reference to fig 5.6.4 we can write

$$X_2(s) = [U(s) - X_1(s)] \left[\frac{2}{s+3} \right]$$

$$X_2(s) (s+3) = 2U(s) - 2X_1(s)$$

$$sX_2(s) + 3X_2(s) = 2U(s) - 2X_1(s)$$

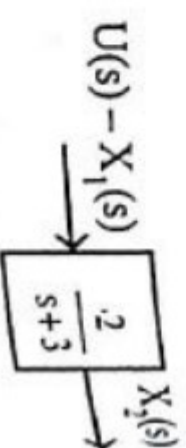


Fig 5.6.4

Problem

On taking inverse Laplace transform

$$\dot{x}_2 + 3x_2 = 2u - 2x_1$$

$$\dot{x}_2 = -2x_1 - 3x_2 + 2u \quad \text{.....(5.6.3)}$$

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From equ(5.6.2) we get, $\dot{x}_1 = x_3$; $\therefore \ddot{x}_1 = \dot{x}_3$

Put $\dot{x}_1 = x_3$ and $\ddot{x}_1 = \dot{x}_3$ in equation (5.6.1)

$$\therefore \dot{x}_3 + x_3 = 2x_2 - 2x_3$$

$$\dot{x}_3 = 2x_2 - 2x_3 - x_3$$

$$\dot{x}_3 = 2x_2 - 3x_3$$

$$\text{.....(5.6.4)}$$

Problem

The state equation are given by equations (5.6.2), (5.6.3) and (5.6.4)

$$\dot{x}_1 = x_3$$

$$\dot{x}_2 = -2x_1 - 3x_2 + 2u$$

$$\dot{x}_3 = 2x_2 - 3x_3$$

The output equation is $y = x_1$

The state model in the matrix form is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u \quad ; \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Problem

KALMAN'S TEST FOR CONTROLLABILITY

$$A^2 = A.A = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 2 & -3 \\ 6 & 9 & -2 \\ -4 & -12 & 9 \end{bmatrix}$$

$$A.B = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -6 \\ 4 \end{bmatrix}$$

$$A^2.B = \begin{bmatrix} 0 & 2 & -3 \\ 6 & 9 & -2 \\ -4 & -12 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \\ -24 \end{bmatrix}$$

Problem

The composite matrix for controllability, $Q_c = [B \quad AB \quad A^2B]$

$$= \begin{bmatrix} 0 & 0 & 4 \\ 2 & -6 & 18 \\ 0 & 4 & -24 \end{bmatrix}$$

$$\text{Determinant of } Q_c = \begin{vmatrix} 0 & 0 & 4 \\ 2 & -6 & 18 \\ 0 & 4 & -24 \end{vmatrix} = 4 \times 8 = 32 ; \text{ Since } |Q_c| \neq 0, \text{ the rank of } Q_c = 3.$$

Hence the system is completely state controllable

Problem

KALMAN'S TEST FOR OBSERVABILITY

$$A^T = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix}^T = \begin{bmatrix} 0 & -2 & 0 \\ 0 & -3 & 2 \\ 1 & 0 & -3 \end{bmatrix}$$

$$C^T = [1 \ 0 \ 0]^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(A^T)^2 = \begin{bmatrix} 0 & -2 & 0 \\ 0 & -3 & 2 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} 0 & -2 & 0 \\ 0 & -3 & 2 \\ 1 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 6 & -4 \\ 2 & 9 & -12 \\ -3 & -2 & 9 \end{bmatrix}$$

$$A^T \dot{C}^T = \begin{bmatrix} 0 & -2 & 0 \\ 0 & -3 & 2 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem

$$(A^T)^2 C^T = \begin{bmatrix} 0 & 6 & -4 \\ 2 & 9 & -12 \\ -3 & -2 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix}$$

The composite matrix for observability $\left\{ Q_o = \begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 1 & -3 \end{bmatrix} \right.$

$$\text{Determinant of } Q_o = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 1 & -3 \end{vmatrix} = 1 \times -2 = -2 ; \text{ Since } |Q_o| \neq 0, \text{ the rank of } Q_o = 3$$

Hence the system is completely observable. (or all the state variables of the system are observable).

Gilbert's Test for Controllability

Gilbert's method of testing controllability

Case(i) : When the system matrix has distinct eigenvalues

In this case the system matrix can be diagonalized and the state model can be converted to canonical form.

Consider the state model of the system,

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

The state model can be converted to canonical form by a transformation, $X = MZ$, where M is the modal matrix and Z is the transformed state variable vector.

The transformed state model is given by

$$\dot{Z} = \Lambda Z + \tilde{B}U$$

$$Y = \tilde{C}Z + DU$$

$$\text{where } \Lambda = M^{-1}AM$$

$$\tilde{B} = M^{-1}B$$

$$\tilde{C} = CM$$

Gilbert's Test for Controllability

In this case the necessary and sufficient condition for complete controllability is that, the matrix $\bar{B}$ must have no rows with all zeros. If any row of the matrix $\bar{B}$ is zero then the corresponding state variable is uncontrollable.

Case(ii) : When the system matrix has repeated eigenvalues

In this case, the system matrix cannot be diagonalized but can be transformed to Jordan canonical form.

Gilbert's Test for Controllability

Consider the state model of the system,

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

The state model can be transformed to Jordan canonical form by a transformation, $X = MZ$, where M is model matrix and Z is the transformed state variable vector.

The transformed state model is given by,

$$\dot{Z} = JZ + \tilde{B}U$$

$$Y = \tilde{C}Z + DU$$

$$\text{where } J = M^{-1}AM$$

$$\tilde{B} = M^{-1}B$$

$$\tilde{C} = CM$$

In this case, the system is completely controllable if the elements of any row of $\tilde{B}$ that correspond to the last row of each Jordan block are not all zero and the rows corresponding to other state variables must not have all zeros.

Gilbert's Test for Observability

Gilbert's method of testing observability

Consider a state model of n^{th} order system, $\dot{X} = AX + BU$; $Y = CX + DU$

The state model can be transformed to a canonical or Jordan canonical form by a transformation, $X = MZ$, where M is the modal matrix and Z is the transformed state variable vector.

The transformed state model is,

$$\begin{aligned}\dot{Z} &= \Lambda Z + \tilde{B}U \\ Y &= \tilde{C}Z + DU\end{aligned}\quad \text{(or)} \quad \begin{aligned}\dot{Z} &= JZ + \tilde{B}U \\ Y &= \tilde{C}Z + DU\end{aligned}$$

where $\Lambda = M^{-1}AM$; if eigenvalues are distinct ; $\tilde{B} = M^{-1}B$

$J = M^{-1}AM$; if eigenvalues have multiplicity ; $\tilde{C} = CM$

The necessary and sufficient condition for complete observability is that none of the columns of the matrix $\tilde{C}$ be zero. If any of the column's of $\tilde{C}$ has all zeros then the corresponding state variable is not observable.

Gilbert's Test

The state model in the matrix form is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u \quad ; \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

To find eigenvalues

Here the system matrix, $A = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix}$

The characteristic equation is $|\lambda I - A| = 0$

$$(\lambda I - A) = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} = \begin{bmatrix} \lambda & 0 & -1 \\ 2 & \lambda + 3 & 0 \\ 0 & -2 & \lambda + 3 \end{bmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda & 0 & -1 \\ 2 & \lambda + 3 & 0 \\ 0 & -2 & \lambda + 3 \end{vmatrix} = \lambda(\lambda + 3)^2 - 1(-4) = \lambda(\lambda^2 + 6\lambda + 9) + 4$$

Gilbert's Test

$$\begin{aligned}
 &= \lambda^3 + 6\lambda^2 + 9\lambda + 4 = (\lambda + 1)(\lambda^2 + 5\lambda + 4) \\
 &= (\lambda + 1)(\lambda + 1)(\lambda + 4) = (\lambda + 1)^2(\lambda + 4)
 \end{aligned}$$

The eigenvalues are $\lambda_1 = -1$, $\lambda_2 = -1$, and $\lambda_3 = -4$,

$\lambda = -1$	1	6	9	4
	↓	-1	-5	-4
	1	5	4	0

Gilbert's Test

To find eigenvectors

$$(\lambda_1 I - A) = \lambda_1 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & -1 \\ 2 & \lambda_1 + 3 & 0 \\ 0 & -2 & \lambda_1 + 3 \end{bmatrix}$$

Let C_{11} , C_{12} and C_{13} be cofactors along 1st row of the matrix $(\lambda_1 I - A)$

$$C_{11} = (+1) \begin{vmatrix} \lambda_1 + 3 & 0 \\ -2 & \lambda_1 + 3 \end{vmatrix} = (\lambda_1 + 3)^2 = \lambda_1^2 + 6\lambda_1 + 9$$

$$C_{12} = (-1) \begin{vmatrix} 2 & 0 \\ 0 & \lambda_1 + 3 \end{vmatrix} = -(2(\lambda_1 + 3)) = -2\lambda_1 - 6$$

$$C_{13} = (+1) \begin{vmatrix} 2 & \lambda_1 + 3 \\ 0 & -2 \end{vmatrix} = -4$$

Gilbert's Test

$$m_1 = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{13} \end{bmatrix} = \begin{bmatrix} \lambda_1^2 + 6\lambda_1 + 9 \\ -2\lambda_1 - 6 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 - 6 + 9 \\ 2 - 6 \\ -4 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ -4 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} \frac{d}{d\lambda_1} C_{11} \\ \frac{d}{d\lambda_1} C_{12} \\ \frac{d}{d\lambda_1} C_{13} \end{bmatrix} = \begin{bmatrix} 2\lambda_1 + 6 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 + 6 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix}$$

$$(\lambda_3 I - A) = (-4) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} = \begin{bmatrix} -4 & 0 & -1 \\ 2 & -1 & 0 \\ 0 & -2 & -1 \end{bmatrix}$$

Gilbert's Test

Let C_{11}, C_{12} and C_{13} be the cofactors along 1st row of the matrix $(\lambda_3 I - A)$.

$$C_{11} = (+1) \begin{vmatrix} -1 & 0 \\ -2 & -1 \end{vmatrix} = 1 ; C_{12} = (-1) \begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix} = 2 ; C_{13} = (+1) \begin{bmatrix} 2 & -1 \\ 0 & -2 \end{bmatrix} = -4$$

$$\therefore m_3 = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{13} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}$$

Gilbert's Test

To find canonical form of state model

The modal matrix, M is given by

$$M = [m_1 \ m_2 \ m_3] = \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix}$$

$$M^{-1} = \frac{[\text{Cofactor of } M]^T}{\text{Determinant of } M} = \frac{M_{\text{cof}}^T}{\Delta_M}$$

$$\Delta_M = \begin{vmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{vmatrix} = 4(8) - 4(24) + 1(-8) = 32 - 96 - 8 = -72$$

$$M_{\text{cof}}^T = \begin{bmatrix} 8 & -24 & -8 \\ 16 & -12 & -16 \\ 10 & -12 & 8 \end{bmatrix}^T = \begin{bmatrix} 8 & 16 & 10 \\ -24 & -12 & -12 \\ -8 & -16 & 8 \end{bmatrix}$$

$$M^{-1} = \frac{1}{-72} \begin{bmatrix} 8 & 16 & 10 \\ -24 & -12 & -12 \\ -8 & -16 & 8 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} -2 & -4 & -2.5 \\ 6 & 3 & 3 \\ 2 & 4 & -2 \end{bmatrix}$$

Gilbert's Test

$$J = M^{-1}AM = \frac{1}{18} \begin{bmatrix} -2 & -4 & -2.5 \\ 6 & 3 & 3 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} 8 & 7 & 5.5 \\ -6 & -3 & -3 \\ -8 & -16 & 8 \end{bmatrix} \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} -18 & 18 & 0 \\ 0 & -18 & 0 \\ 0 & 0 & -72 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -4 \end{bmatrix} \quad \swarrow \text{Jordan block}$$

$$\tilde{B} = M^{-1}B = \frac{1}{18} \begin{bmatrix} -2 & -4 & -2.5 \\ 6 & 3 & 3 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -8/18 \\ 6/18 \\ 8/18 \end{bmatrix} = \begin{bmatrix} -4/9 \\ 3/9 \\ 4/9 \end{bmatrix}$$

Gilbert's Test

$$\tilde{C} = CM = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 1 \end{bmatrix}$$

The Jordan canonical form of state model is shown below.

$$\dot{Z} = JZ + \tilde{B}U ; \quad Y = \tilde{C}Z + DU \quad (\text{Here } DU \text{ is not defined})$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} -4/9 \\ 3/9 \\ 4/9 \end{bmatrix} [u] ; \quad Y = \begin{bmatrix} 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

CONCLUSION

It is observed that the elements of the rows of $\tilde{B}$ are not all zeros. Hence the system is completely controllable. (or state controllable).

It is observed that the elements of the columns of $\tilde{C}$ are not all zeros. Hence the system is completely observable [i.e, all the state variables are observable].

PBH Test for Controllability

PBH Tests

- PBH Test of Controllability

Pair (A, B) is controllable if and only if for $\lambda \in \sigma(A)$

$$\text{rank} [A - \lambda I \quad B] = n$$

- PBH Test for Observability

Pair (A, C) is observable if and only if for $\lambda \in \sigma(A)$

$$\text{rank} [A' - \lambda I \quad C'] = n$$

PBH Test for Controllability

Theorem 2.5 (PBH Test):

(A, B) is controllable if and only if $\text{Rank}[A - \lambda I \quad B] = n$ for all eigenvalues λ of A .

Proof: The statement is equivalent to:

(A, B) is not controllable if and only if $\text{rank}[A - \lambda I \quad B] < n$ for some eigenvalue λ of A .

(i) We first prove $\text{rank}[A - \lambda I \quad B] < n$ implies (A, B) is not controllable. Suppose $\text{rank}[A - \lambda I \quad B] < n$ for some eigenvalue λ , possibly complex. Thus there exists a vector x , possibly complex such that

$$x^*[A - \lambda I \quad B] = 0$$

where x^* is complex conjugate transpose of x . This results in

PBH Test for Controllability

Example 1: Consider the following (A, B) pair

$$A = \begin{bmatrix} -3 & 2 \\ 1 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This corresponds to the bridge circuit example with the following parameter values: $\alpha_1 = \alpha_2 = 1$, $\beta_1 = 2$, and $\beta_2 = 1$. We know that since $\alpha_1 = \alpha_2$, this system is not controllable. We can also verify that the controllability matrix is

$$\mathcal{C}_{AB} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

which is singular. Let us apply the PBH test. First $\det(sI - A) = s^2 + 5s + 4$ so that the eigenvalues are -4 and -1 . For the eigenvalue -1 , PBH test gives

$$[A - (-1)I \quad B] = \begin{bmatrix} -2 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

which has rank 2. On the other hand, for the eigenvalue -4 , PHB test gives

$$[A - (-4)I \quad B] = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

which has rank 1. By the PBH test, the system is not controllable and -4 is an uncontrollable eigenvalue.

Duality Property

Duality Property

The concepts of controllability and observability are dual concepts and it is proposed by Kalman as principle of duality.

The principle of duality states that a system is completely state controllable if and only if its dual system is completely observable or viceversa. [i.e., if the system is observable then its dual is controllable]. Using the principle of duality, the observability of a given system can be checked by testing the state controllability of its dual or vice-versa.

Consider the system S_1 , described by the state model shown below.

$$\dot{X} = AX + BU$$

$$Y = CX$$

Let the dual of system S_1 be denoted as S_2 and the dual system S_2 is described by the following state model.

$$\dot{Z} = A^T Z + C^T V$$

$$N = B^T Z$$

Duality Property

Consider the system S_1 , described by the state model shown below.

$$\dot{X} = AX + BU$$

$$Y = CX$$

Let the dual of system S_1 be denoted as S_2 and the dual system S_2 is described by the following state model.

$$\dot{Z} = A^T Z + C^T V$$

$$N = B^T Z$$

where, Z = State vector of dual system

V = Input vector of dual system

; N = Output vector of dual system

For the system S_1 , the composite matrix, Q_{c1} for controllability is given by equ(5.66) and the composite matrix, Q_{o1} for observability is given by equ(5.67).

$$Q_{c1} = [B \quad AB \quad A^2B \quad \dots \dots \dots A^{n-1}B] \quad \dots \dots (5.66)$$

$$Q_{o1} = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad \dots \dots \dots (A^T)^{n-1} C^T] \quad \dots \dots (5.67)$$

Duality Property

For the dual system S_2 the composite matrix, Q_{c2} for controllability is given by equ(5.68) and the composite matrix Q_{o2} for observability is given by equ(5.69)

$$Q_{c2} = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad \dots \dots \dots (A^T)^{n-1} C^T] \quad \dots \dots (5.68)$$

$$Q_{o2} = [A \quad AB \quad A^2B \quad \dots \dots \dots A^{n-1} B] \quad \dots \dots (5.69)$$

From equations (5.66) and (5.69) we get $Q_{c1} = Q_{o2}$, hence if the system S_1 is controllable then its dual system S_2 is observable.

From equations (5.67) and (5.68) we get $Q_{o1} = Q_{c2}$, hence if the system S_1 is observable then its dual system S_2 is controllable.

Controller Design in State Space by Pole Placement

Conventional method of controller design

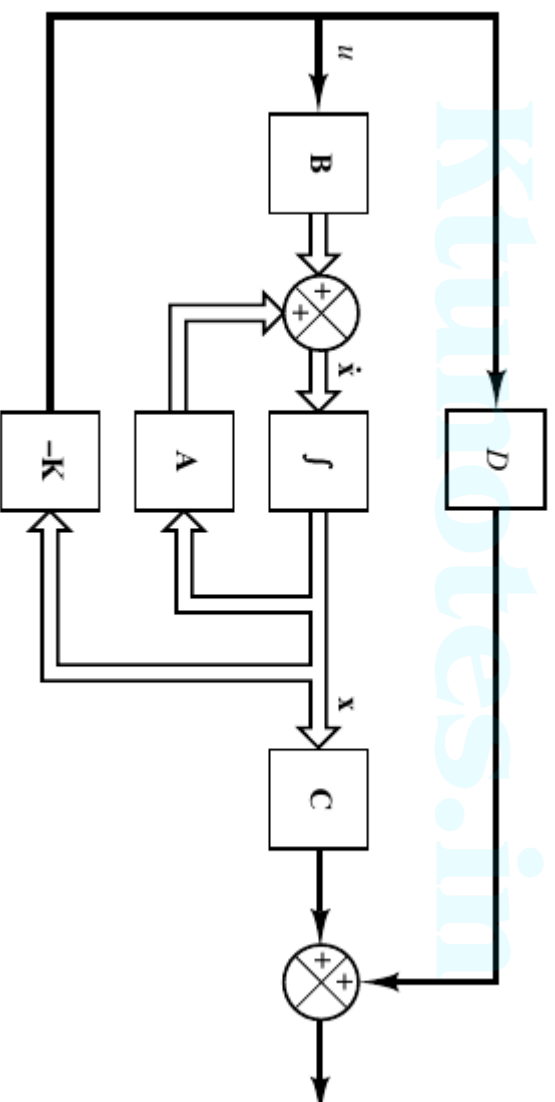
- We have already studied this method in Module I & II
- **Conventional Design**: In the conventional approach to the design of a single-input, single-output control system, we design a controller (compensator) such that the **dominant closed-loop poles** have a desired damping ratio ζ and a desired undamped natural frequency ω_n . In this approach, the order of the system may be raised by 1 or 2 unless pole-zero cancellation takes place. Note that in this approach we assume the effects on the responses of non-dominant closed-loop poles to be negligible.

Controller Design in State Space by Pole Placement

- **Pole Placement Design:** In this section we shall present a design method commonly called the **pole-placement** or **pole-assignment** technique. We assume that all state variables are measurable and are available for feedback. It will be shown that if the system considered is completely state **controllable**, then poles of the closed-loop system may be placed at any desired locations by means of state feedback through an appropriate **state feedback gain matrix [K]**.
- Different from specifying only dominant closed-loop poles (the conventional design approach), the present **pole-placement** approach specifies all closed-loop poles.

State Feedback Controller

- Figure shows the system with state feedback controller. The gain K is the state feedback gain matrix. The control law is $u = -Kx$



State Feedback Controller

Consider a control system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

We shall choose the control signal to be

$$u = -\mathbf{K}\mathbf{x}$$

Then

$$\dot{x} = Ax - BKx$$

Or

$$\dot{x} = (A - BK)x$$

State Feedback Controller

The solution of this equation is given by

$$x(t) = e^{(A-BK)t} x(0)$$

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Design of State Feedback Controller

Method I

1. Determine the characteristic polynomial of original system. The characteristic polynomial is given by $|\lambda I - A| = 0$

$$\text{Let, } |\lambda I - A| = \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_{n-1} \lambda + a_n.$$

2. Determine the desired characteristic polynomial from the specified closed loop poles.

Let the specified or desired closed loop poles be $\mu_1, \mu_2, \mu_3, \dots, \mu_n$.

Now the desired characteristic polynomial is given by

$$(\lambda - \mu_1)(\lambda - \mu_2)(\lambda - \mu_3) \dots (\lambda - \mu_n) = \lambda^n + \alpha_1 \lambda^{n-1} + \alpha_2 \lambda^{n-2} + \dots + \alpha_{n-1} \lambda + \alpha_n.$$

Design of State Feedback Controller

Method I

3. Determine the transformation matrix, P_c which transforms the original state model to controllable phase variable form.

The transformation matrix, $P_c = \begin{bmatrix} P_1 \\ P_1 A \\ \vdots \\ P_1 A^{n-1} \end{bmatrix}$

and, $P_1 = [0 \ 0 \ \dots \ 0 \ 1] Q_c^{-1}$

4. Determine the state feedback gain matrix, from the following equation.

$$K = [\alpha_n - a_n \quad \alpha_{n-1} - a_{n-1} \quad \dots \quad \alpha_2 - a_2 \quad \alpha_1 - a_1] P_c^{-1}.$$

Design of State Feedback Controller

Method II

1. Determine the characteristic polynomial of the system with state feedback, which is given by, $|\lambda I - (A - BK)| = 0$.

Here take, $K = [k_1 \ k_2 \ k_3 \ \dots \ k_n]$

Let $|\lambda I - (A - BK)| = |\lambda I - A + BK| = \lambda^n + b_1 \lambda^{n-1} + b_2 \lambda^{n-2} + \dots + b_{n-1} \lambda + b_n$.

The coefficients of this polynomial $b_1, b_2, b_3, \dots, b_n$ will be functions of $k_1, k_2, k_3, \dots, k_n$.

2. Determine the desired characteristic polynomial from the specified closed loop poles.

Let the specified or desired closed loop poles be $\mu_1, \mu_2, \mu_3, \dots, \mu_n$.

Now the desired characteristic polynomial is given by,

$$(\lambda - \mu_1)(\lambda - \mu_2)(\lambda - \mu_3) \dots (\lambda - \mu_n) = \lambda^n + \alpha_1 \lambda^{n-1} + \alpha_2 \lambda^{n-2} + \dots + \alpha_{n-1} \lambda + \alpha_n.$$

3. By equating the coefficients of polynomials obtained in step-1 and step-2, we get n-number of equations.

$$\text{i.e, } b_1 = \alpha_1 ; b_2 = \alpha_2 ; \dots ; b_{n-1} = \alpha_{n-1} \text{ and } b_n = \alpha_n.$$

On solving these equations we get the elements $k_1, k_2, \dots, k_n$ of state feedback gain matrix, K.

Problem

- Problem1

A single-input system is described by the following state equation.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix} u$$

Design a state feedback controller which will give closed-loop poles at $-1 \pm j2$, -6 .

Problem

- Solution

Check for controllability

$$\text{Given that } A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix}$$

$$A^2 = A.A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -10 \\ 8 \\ 21 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix} \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ -26 \\ -75 \end{bmatrix}$$

Problem

- Solution

$$\text{Composite matrix } \left\{ \begin{array}{l} Q_c = [B \quad AB \quad A^2B] \\ \text{for controllability} \end{array} \right\} = \begin{bmatrix} 10 & -10 & 10 \\ 1 & 8 & -26 \\ 0 & 21 & -75 \end{bmatrix}$$

$$\text{Determinant of } Q_c = \Delta_{qc} = \begin{vmatrix} 10 & -10 & 10 \\ 1 & 8 & -26 \\ 0 & 21 & -75 \end{vmatrix}$$

$$\begin{aligned} \therefore \Delta_{qc} &= 10 [8 \times (-75) - 21 \times (-26)] + 10 [-75] + 10 [21] \\ &= -540 - 750 + 210 \\ &= -1080 \end{aligned}$$

Since, $\Delta_{qc} \neq 0$, The system is completely state controllable.

Problem

- State feedback controller

To find Q_c^{-1}

From equations (5.9.2) and (5.9.3) we get,

$$Q_c = \begin{bmatrix} 10 & -10 & 10 \\ 1 & 8 & -26 \\ 0 & 21 & -75 \end{bmatrix} \text{ and } \Delta_{qc} = -1080.$$

$$Q_c^{-1} = \frac{[\text{cofactor of } Q_c]^T}{\text{Determinant of } Q_c} = \frac{1}{\Delta_{qc}} \begin{bmatrix} -54 & 75 & 21 \\ -540 & -750 & -210 \\ 180 & 270 & 90 \end{bmatrix}^T$$

$$= \frac{1}{-1080} \begin{bmatrix} -54 & -540 & 180 \\ 75 & -750 & 270 \\ 21 & -210 & 90 \end{bmatrix} = \begin{bmatrix} 0.05 & 0.5 & -0.1667 \\ -0.0694 & 0.6944 & -0.25 \\ -0.0194 & 0.1944 & -0.0833 \end{bmatrix}$$

Problem

- Characteristic Polynomial

To find desired characteristic polynomial

The desired closed loop poles are,

$$\mu_1 = -1 + j2, \mu_2 = -1 - j2 \text{ and } \mu_3 = -6$$

Hence the desired characteristic polynomial is,

$$\begin{aligned}(\lambda - \mu_1)(\lambda - \mu_2)(\lambda - \mu_3) &= (\lambda + 1 - j2)(\lambda + 1 + j2)(\lambda + 6) \\&= ((\lambda + 1)^2 - (j2)^2)(\lambda + 6) \\&= (\lambda^2 + 2\lambda + 1 + 4)(\lambda + 6) \\&= (\lambda^2 + 2\lambda + 5)(\lambda + 6) \\&= \lambda^3 + 2\lambda^2 + 5\lambda + 6\lambda^2 + 12\lambda + 30 \\&= \lambda^3 + 8\lambda^2 + 17\lambda + 30\end{aligned}$$

The desired characteristic polynomial is $\lambda^3 + 8\lambda^2 + 17\lambda + 30 = 0$

Problem

- State variable feedback matrix

To determine the state variable feedback matrix, K

Method - I

The characteristic equation of original system is given by,

$$[\lambda I - A] = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} = \begin{bmatrix} \lambda+1 & 0 & 0 \\ -1 & \lambda+2 & 0 \\ -2 & -1 & \lambda+3 \end{bmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda+1 & 0 & 0 \\ -1 & \lambda+2 & 0 \\ -2 & -1 & \lambda+3 \end{vmatrix} = (\lambda+1)(\lambda+2)(\lambda+3)$$

$$\begin{aligned} &= (\lambda+1)(\lambda^2+5\lambda+6) = \lambda^3+5\lambda^2+6\lambda+\lambda^2+5\lambda+6 \\ &= \lambda^3+6\lambda^2+11\lambda+6 \end{aligned}$$

The characteristic polynomial of original system is,

$$\lambda^3+6\lambda^2+11\lambda+6=0$$

From equation (5.9.5) we get the desired characteristic polynomial as

$$\lambda^3+8\lambda^2+17\lambda+10=0$$

.....(5.9.6)

.....(5.9.7)

Problem

• K

$$Q_c^{-1} = \begin{bmatrix} 0.05 & 0.5 & -0.1667 \\ -0.0694 & 0.6944 & -0.25 \\ -0.0194 & 0.1944 & -0.0833 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} Q_c^{-1} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.05 & 0.5 & -0.1667 \\ -0.0694 & 0.6944 & -0.25 \\ -0.0194 & 0.1944 & -0.0833 \end{bmatrix}$$

$$= \begin{bmatrix} -0.0194 & 0.1944 & -0.0833 \end{bmatrix}$$

$$P_1 A = \begin{bmatrix} -0.0194 & 0.1944 & -0.0833 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0472 & -0.4721 & 0.2499 \end{bmatrix}$$

Problem

• K

$$P_1 A^2 = \begin{bmatrix} -0.0194 & 0.1944 & -0.0833 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -0.0195 & 1.1941 & -0.7497 \end{bmatrix}$$

$$P_c = \begin{bmatrix} P_1 \\ P_1 A \\ P_1 A^2 \end{bmatrix} = \begin{bmatrix} -0.0194 & 0.1944 & -0.0833 \\ 0.0472 & -0.4721 & 0.2499 \\ -0.0195 & 1.1941 & -0.7497 \end{bmatrix}$$

The state feedback gain matrix, $K = [\alpha_3 - a_3 \quad \alpha_2 - a_2 \quad \alpha_1 - a_1] P_c$

From equation (5.9.7) we get, $\alpha_3 = 30$; $\alpha_2 = 17$; $\alpha_1 = 8$

From equation (5.9.6) we get, $a_3 = 6$; $a_2 = 11$; $a_1 = 6$

$$\therefore K = \begin{bmatrix} 30-6 & 17-11 & 8-6 \end{bmatrix} P_c$$

$$= \begin{bmatrix} 24 & 6 & 2 \end{bmatrix} \begin{bmatrix} -0.0194 & 0.1944 & -0.0833 \\ 0.0472 & -0.4721 & 0.2499 \\ -0.0195 & 1.1941 & -0.7497 \end{bmatrix}$$

$$= \begin{bmatrix} -0.22 & 4.22 & -2 \end{bmatrix}$$

Problem

- Method - II

From the given state model we get

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{Let, } K = [k_1 \quad k_2 \quad k_3]$$

The characteristic polynomial of the systems with state feedback is given by,

$$|\lambda I - (A - BK)| = |\lambda I - A + BK| = 0$$

$$[\lambda I - A + BK] = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} + \begin{bmatrix} 10 \\ 1 \\ 0 \end{bmatrix} [k_1 \quad k_2 \quad k_3]$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} + \begin{bmatrix} 10k_1 & 10k_2 & 10k_3 \\ k_1 & k_2 & k_3 \\ 0 & 0 & 0 \end{bmatrix}$$

Problem

- Method II

$$= \begin{bmatrix} \lambda + 1 + 10k_1 & 10k_2 & 10k_3 \\ -1 + k_1 & \lambda + 2 + k_2 & k_3 \\ -2 & -1 & \lambda + 3 \end{bmatrix}$$

$$|\lambda I - A + BK| = \begin{vmatrix} \lambda + 1 + 10k_1 & 10k_2 & 10k_3 \\ -1 + k_1 & \lambda + 2 + k_2 & k_3 \\ -2 & -1 & \lambda + 3 \end{vmatrix}$$

$$= (\lambda + 1 + 10k_1) [(\lambda + 2 + k_2)(\lambda + 3) + k_3] - 10k_2 [(-1 + k_1)(\lambda + 3) + 2k_3] + 10k_3 [-(-1 + k_1) + 2(\lambda + 2 + k_2)]$$

$$= [\lambda + (1 + 10k_1)] [\lambda^2 + 3\lambda + 2\lambda + 6 + \lambda k_2 + 3k_2 + k_3] - 10k_2 [-\lambda - 3 + \lambda k_1 + 3k_1 + 2k_3] + 10k_3 [1 - k_1 + 2\lambda + 4 + 2k_2]$$

Problem

- Method II

$$= [\lambda + (1 + 10k_1)] [\lambda^2 + (5 + k_2)\lambda + (6 + 3k_2 + k_3)] \\ - 10k_2 [(-1 + k_1)\lambda + (-3 + 3k_1 + 2k_3)] + 10k_3 [2\lambda + (5 - k_1 + 2k_2)]$$

$$= \lambda^3 + (5 + k_2)\lambda^2 + (6 + 3k_2 + k_3)\lambda \\ + (1 + 10k_1)\lambda^2 + (1 + 10k_1)(5 + k_2)\lambda + (1 + 10k_1)(6 + 3k_2 + k_3) \\ - 10k_2(-1 + k_1)\lambda - 10k_2(-3 + 3k_1 + 2k_3) + 20k_3\lambda + 10k_3(5 - k_1 + 2k_2)$$

$$= \lambda^3 + (6 + 10k_1 + k_2)\lambda^2 + (6 + 3k_2 + k_3 + 5 + k_2 + 50k_1 + 10k_1k_2 \\ + 10k_2 - 10k_1k_2 + 20k_3)\lambda + (6 + 3k_2 + k_3 + 60k_1 + 30k_1k_2 \\ + 10k_1k_3 + 30k_2 - 30k_1k_2 - 20k_2k_3 + 50k_3 - 10k_1k_3 + 20k_2k_3)$$

$$= \lambda^3 + (6 + 10k_1 + k_2)\lambda^2 + (11 + 50k_1 + 14k_2 + 21k_3)\lambda \\ + (6 + 60k_1 + 33k_2 + 51k_3)$$

Problem

- Me

The characteristic polynomial of system with state feedback is

$$\lambda^3 + (6 + 10k_1 + k_2)\lambda^2 + (11 + 50k_1 + 14k_2 + 21k_3)\lambda + (6 + 60k_1 + 33k_2 + 51k_3) = 0 \quad \dots(5.9.8)$$

From equation (5.9.5) we get the desired characteristic polynomial as,

$$\lambda^3 + 8\lambda^2 + 17\lambda + 30 = 0 \quad \dots\dots(5.9.9)$$

On equating the coefficients of λ^2 term in equations (5.9.8) and (5.9.9) we get,

$$6 + 10k_1 + k_2 = 8$$

$$10k_1 + k_2 = 8 - 6$$

$$\therefore 10k_1 + k_2 = 2 \quad \dots\dots(5.9.10)$$

On equating the coefficients of λ^1 term in equations (5.9.8) and (5.9.9) we get,

$$11 + 50k_1 + 14k_2 + 21k_3 = 17$$

$$50k_1 + 14k_2 + 21k_3 = 17 - 11$$

$$\therefore 50k_1 + 14k_2 + 21k_3 = 6 \quad \dots\dots(5.9.11)$$

On equating the coefficients of λ^0 term (constant) in equations (5.9.8) and (5.9.9) we

Problem

- Method II

$$6 + 60k_1 + 33k_2 + 51k_3 = 30$$

$$60k_1 + 33k_2 + 51k_3 = 30 - 6$$

$$\therefore 60k_1 + 33k_2 + 51k_3 = 24$$

.....(5.9.12)

The equations (5.9.10), (5.9.11) and (5.9.12) can be arranged in the matrix form and k_1 , k_2 and k_3 are solved using cramer's rule as shown below.

$$\begin{bmatrix} 10 & 1 & 0 \\ 50 & 14 & 21 \\ 60 & 33 & 51 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 24 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 10 & 1 & 0 \\ 50 & 14 & 21 \\ 60 & 33 & 51 \end{vmatrix} = 10(14 \times 51 - 33 \times 21) - 1(50 \times 51 - 60 \times 21) \\ = 210 - 1290 = -1080$$

Problem

- Method II

$$\Delta_1 = \begin{vmatrix} 2 & 1 & 0 \\ 6 & 14 & 21 \\ 24 & 33 & 51 \end{vmatrix} = 2(14 \times 51 - 33 \times 21) - 1(6 \times 51 - 24 \times 21) = 42 + 198 = 240$$

$$\Delta_2 = \begin{vmatrix} 10 & 2 & 0 \\ 50 & 6 & 21 \\ 60 & 24 & 51 \end{vmatrix} = 10(6 \times 51 - 24 \times 21) - 2(50 \times 51 - 60 \times 21) = -1980 - 2580 = -4560$$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 10 & 1 & 2 \\ 50 & 14 & 6 \\ 60 & 33 & 24 \end{vmatrix} = 10(14 \times 24 - 33 \times 6) - 1(50 \times 24 - 60 \times 6) \\ &\quad + 2(50 \times 33 - 60 \times 14) \\ &= 1380 - 840 + 1620 = 2160 \end{aligned}$$

Problem

- Method II

$$k_1 = \frac{\Delta_1}{\Delta} = \frac{240}{-1080} = -0.22$$

$$k_2 = \frac{\Delta_2}{\Delta} = \frac{-4560}{-1080} = 4.22$$

$$k_3 = \frac{\Delta_3}{\Delta} = \frac{2160}{-1080} = -2$$

The state feedback gain matrix, $K = [k_1 \quad k_2 \quad k_3] = [-0.22 \quad 4.22 \quad -2]$

Problem

- Method III

Method III

From equation (5.9.5) we get the desired characteristic polynomial as,

$$\lambda^3 + 8\lambda^2 + 17\lambda + 30 = 0$$

.....(5.9.13)

Here, $\phi(A) = A^3 + \alpha_1 A^2 + \alpha_2 A + \alpha_3 I$

From equation (5.9.13) we get, $\alpha_1 = 8$; $\alpha_2 = 17$; $\alpha_3 = 30$

From the given state equation and equation (5.9.1) we get,

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} \text{ and } A^2 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix}$$

Problem

• Method III

$$\therefore A^3 = A^2 \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 7 & -8 & 0 \\ 20 & 19 & -27 \end{bmatrix}$$

$$\therefore \phi(A) = A^3 + \alpha_1 A^2 + \alpha_2 A + \alpha_3 I$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 7 & -8 & 0 \\ 20 & 19 & -27 \end{bmatrix} + 8 \begin{bmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ -7 & -5 & 9 \end{bmatrix} + 17 \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & -3 \end{bmatrix} + 30 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 7 & -8 & 0 \\ 20 & 19 & -27 \end{bmatrix} + \begin{bmatrix} 8 & 0 & 0 \\ -24 & 32 & 0 \\ -56 & -40 & 72 \end{bmatrix} + \begin{bmatrix} -17 & 0 & 0 \\ 17 & -34 & 0 \\ 34 & 17 & -51 \end{bmatrix} + \begin{bmatrix} 30 & 0 & 0 \\ 0 & 30 & 0 \\ 0 & 0 & 30 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ -2 & -4 & 24 \end{bmatrix}$$

Problem

- Method III

From equation (5.9.4) we get,

$$Q_c^{-1} = \begin{bmatrix} 0.05 & 0.5 & -0.1667 \\ -0.0694 & 0.6944 & -0.25 \\ -0.0194 & 0.1944 & -0.0833 \end{bmatrix}$$

From Ackermann's formula we get,

$$K = [0 \ 0 \ 1] Q_c^{-1} \phi(A)$$

$$= [0 \ 0 \ 1] \begin{bmatrix} 0.05 & 0.5 & -0.1667 \\ -0.0694 & 0.6944 & -0.25 \\ -0.0194 & 0.1944 & -0.0833 \end{bmatrix} \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ -2 & -4 & 24 \end{bmatrix}$$

$$= [-0.0194 \ 0.1944 \ -0.0833] \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ -2 & -4 & 24 \end{bmatrix} = [-0.22 \ 4.22 \ -2]$$

The state feedback gain matrix, $K = [-0.22 \ 4.22 \ -2]$

Problem

- Use any one method of the above

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Problem 2

Consider a linear system described by the transfer function

$$\frac{Y(s)}{U(s)} = \frac{10}{s(s+1)(s+2)}$$

Design a feedback controller with a state feedback so that the closed loop poles are placed at $-2, -1 \pm j1$

Problem 2

SOLUTION

To determine the state equation of the system

$$\text{Given that, } \frac{Y(s)}{U(s)} = \frac{10}{s(s+1)(s+2)} \quad \dots\dots(5.8.1)$$

On cross multiplying the equation (5.8.1) we get,

$$Y(s) [s(s+1)(s+2)] = 10 U(s)$$

$$Y(s) [s(s^2+3s+2)] = 10 U(s)$$

$$Y(s) [s^3+3s^2+2s] = 10 U(s)$$

$$\therefore s^3 Y(s) + 3s^2 Y(s) + 2s Y(s) = 10 U(s) \quad \dots\dots(5.8.2)$$

On taking inverse laplace transform of equation (5.8.2) we get,

$$\ddot{y} + 3\ddot{y} + 2\dot{y} = 10u \quad \dots\dots(5.8.3)$$

Problem 2

Let us define state variables as follows,

$$x_1 = y ; x_2 = \dot{y} ; x_3 = \ddot{y}$$

Put $\ddot{y} = \dot{x}_3$; $\dot{y} = x_3$; $\dot{y} = x_2$ and $y = x_1$ in equation(5.8.3)

$$\therefore \dot{x}_3 + 3x_3 + 2x_2 = 10 u$$

$$\text{or } \dot{x}_3 = -2x_2 - 3x_3 + 10 u$$

The state equations governing the system are

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -2x_2 - 3x_3 + 10 u$$

Problem 2

The state equation in the matrix form is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$$

Check for controllability

$$\text{Given that, } A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}$$

$$A^2 = A.A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & -3 \\ 0 & 6 & 7 \end{bmatrix}$$

Problem 2

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ -30 \end{bmatrix} \quad \text{[} P_c \text{]}$$

$$A^2B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & -3 \\ 0 & 6 & 7 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 10 \\ -30 \\ 70 \end{bmatrix}$$

Composite matrix for controllability, $Q_c = [B \ AB \ A^2B] = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 10 & -30 \\ 10 & -30 & 70 \end{bmatrix}$

Determinant of $Q_c = \Delta_{Q_c} = \begin{vmatrix} 0 & 0 & 10 \\ 0 & 10 & -30 \\ 10 & -30 & 70 \end{vmatrix} = 10(-10 \times 10) = -1000$

Problem 2

To find Q_c^{-1}

• From equation (5.8.6) and (5.8.7) we get

$$Q_c = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 10 & -30 \\ 10 & -30 & 70 \end{bmatrix} \text{ and } \Delta_{Q_c} = -1000.$$

$$Q_c^{-1} = \frac{[\text{cofactor of } Q_c]^T}{\text{Determinant of } Q_c} = \frac{1}{\Delta_{Q_c}} \begin{bmatrix} 610 & -300 & -100 \\ 300 & -100 & 0 \\ -100 & 0 & 0 \end{bmatrix}^T$$

$$= \frac{1}{-1000} \begin{bmatrix} 610 & 300 & -100 \\ -300 & -100 & 0 \\ -100 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -0.61 & 0.3 & 0.1 \\ -0.3 & 0.1 & 0 \\ 0.1 & 0 & 0 \end{bmatrix}$$

Problem 2

The desired closed loop poles are

$$\mu_1 = -2, \mu_2 = -1 + j1 \text{ and } \mu_3 = -1 - j1$$

Hence the desired characteristic polynomial is

$$\begin{aligned}(\lambda - \mu_1)(\lambda - \mu_2)(\lambda - \mu_3) &= (\lambda + 2)(\lambda + 1 - j1)(\lambda + 1 + j1) \\&= (\lambda + 2)\left((\lambda + 1)^2 - (j1)^2\right) \\&= (\lambda + 2)(\lambda^2 + 2\lambda + 1 + 1) \\&= (\lambda + 2)(\lambda^2 + 2\lambda + 2) \\&= \lambda^3 + 2\lambda^2 + 2\lambda + 2\lambda^2 + 4\lambda + 4 \\&= \lambda^3 + 4\lambda^2 + 6\lambda + 4\end{aligned}$$

Problem 2

To determine the state variable feedback matrix, K

Method - I

The characteristic polynomial of original system is given by $|\lambda I - A| = 0$

$$[\lambda I - A] = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 0 & 2 & \lambda + 3 \end{bmatrix}$$

$P = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
 $Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 0 & 2 & \lambda + 3 \end{vmatrix} = \lambda [\lambda (\lambda + 3) + 2] = \lambda^2 (\lambda + 3) + 2\lambda = \lambda^3 + 3\lambda^2 + 2\lambda$$

Problem 2

The characteristic polynomial of original system is,

$$\lambda^3 + 3\lambda^2 + 2\lambda = 0$$

From equ(5.8.9) we get the desired characteristic polynomial as

$$\lambda^3 + 4\lambda^2 + 6\lambda + 4 = 0$$

From equation (5.8.8) we get,

$$Q_c^{-1} = \begin{bmatrix} -0.61 & 0.3 & 0.1 \\ -0.3 & 0.1 & 0 \\ 0.1 & 0 & 0 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} Q_c^{-1} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.61 & 0.3 & 0.1 \\ -0.3 & 0.1 & 0 \\ 0.1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.1 & 0 & 0 \end{bmatrix}$$

Problem 2

$$P_1 A = \begin{bmatrix} 0 & 1 & 0 \\ 0.1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0.1 & 0 \end{bmatrix}$$

$$P_1 A^2 = \begin{bmatrix} 0.1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & -3 \\ 0 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.1 \end{bmatrix}$$

$$\therefore P_c = \begin{bmatrix} P_1 \\ P_1 A \\ P_1 A^2 \end{bmatrix} = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}$$

The state feedback gain matrix, $K = [\alpha_3 - a_3 \quad \alpha_2 - a_2 \quad \alpha_1 - a_1] P_c$

From equation (5.8.11) we get, $\alpha_3 = 4$; $\alpha_2 = 6$; $\alpha_1 = 4$

From equation (5.8.10) we get, $a_3 = 0$; $a_2 = 2$; $a_1 = 3$

Problem 2

$$\therefore K = [4 \ -0 \ 6 \ -2 \ 4 \ -3] P_e$$

$$= [4 \ 4 \ 1] \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} = [0.4 \ 0.4 \ 0.1]$$

THANK YOU

Advanced Control Systems

Module III

Part II

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Syllabus

3 State Feedback Controller Design		(6 hours)
3.1	Controllability & observability: Kalman's, Gilbert's and PBH tests- Duality property	2
3.2	State feedback controller design: State feed-back design via pole placement technique	2
3.3	State observers for LTI systems- Full order and reduced order observers- Design of full order observer design	2
4	Nonlinear Systems	(7 hours)
4.1	Types of nonlinear systems- characteristics of nonlinear systems- peculiar features like Jump resonance, Limit cycles and Frequency entrainment	2
4.2	Describing function Method: Analysis through harmonic linearisation	1
4.3	Determination of describing function of nonlinearities	2
4.4	Application of describing function for stability analysis of autonomous system with single nonlinearity (relay, dead zone and saturation only).	2

State Observers for LTI Systems

In the pole-placement approach to the design of control systems, we assumed that all state variables are available for feedback. In practice, however, not all state variables are available for feedback. Then we need to estimate unavailable state variables.

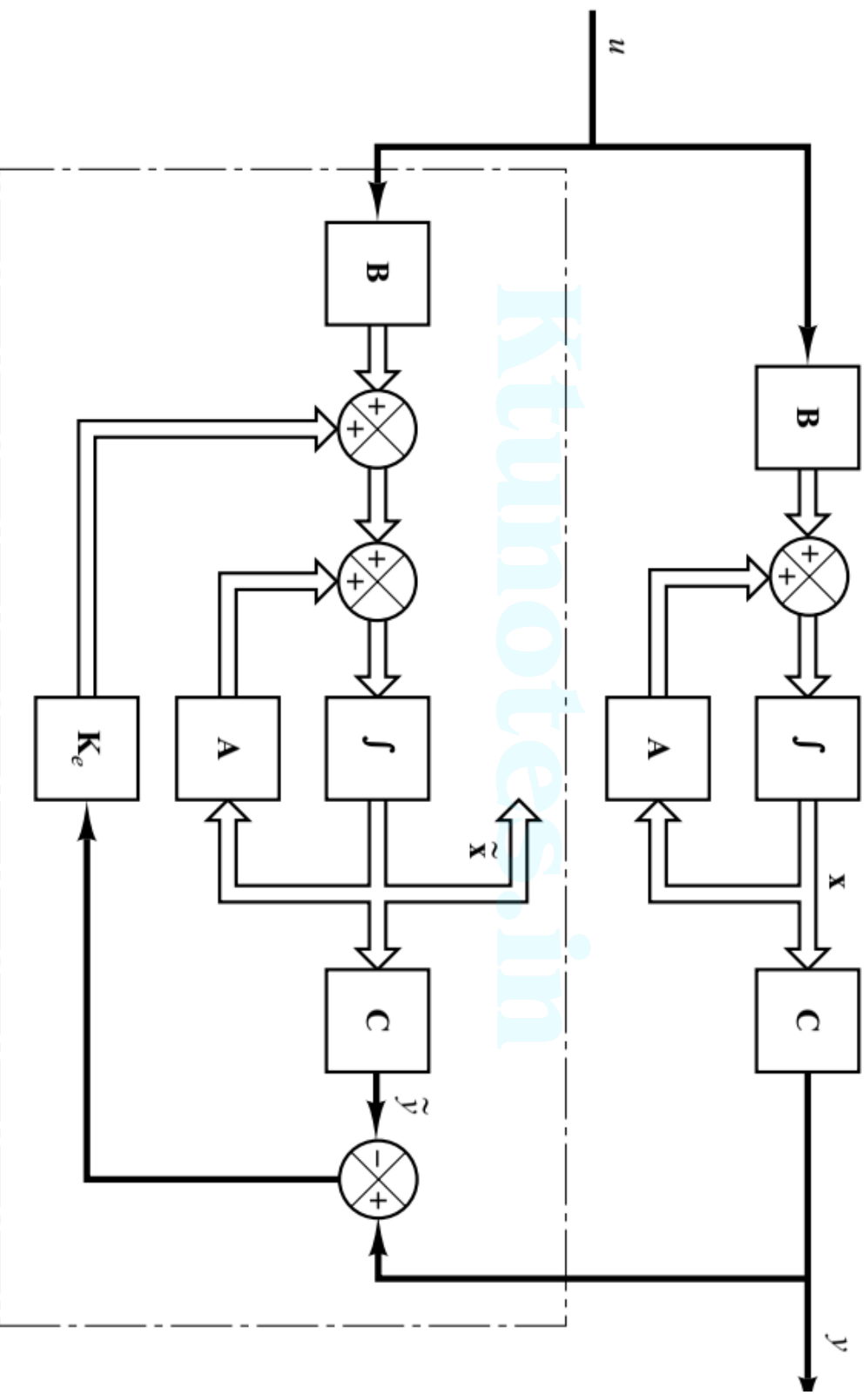
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State observers

Estimation of unmeasurable state variables is commonly called *observation*. A device (or a computer program) that estimates or observes the state variables is called a *state observer*, or simply an *observer*. If the state observer observes all state variables of the system, regardless of whether some state variables are available for direct measurement, it is called a *full-order state observer*. There are times when this will not be necessary, when we will need observation of only the unmeasurable state variables, but not of those that are directly measurable as well. For example, since the output variables are observable and they are linearly related to the state variables, we need not observe all state variables, but observe only $n - m$ state variables, where n is the dimension of the state vector and m is the dimension of the output vector.

An observer that estimates fewer than n state variables, where n is the dimension of the state vector, is called a *reduced-order state observer* or, simply, a *reduced-order observer*. If the order of the reduced-order state observer is the minimum possible, the observer is called a *minimum-order state observer* or *minimum-order observer*. In this section, we shall discuss both the full-order state observer and the minimum-order state observer.

Full order State observer



Full-order state observer

Full order State observer

State Observer. A state observer estimates the state variables based on the measurements of the output and control variables. Here the concept of observability discussed in Section 9–7 plays an important role. As we shall see later, state observers can be designed if and only if the observability condition is satisfied.

In the following discussions of state observers, we shall use the notation $\tilde{\mathbf{x}}$ to designate the observed state vector. In many practical cases, the observed state vector $\tilde{\mathbf{x}}$ is used in the state feedback to generate the desired control vector.

Consider the plant defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (10-55)$$

$$y = \mathbf{C}\mathbf{x} \quad (10-56)$$

The observer is a subsystem to reconstruct the state vector of the plant. The mathematical model of the observer is basically the same as that of the plant, except that we include an additional term that includes the estimation error to compensate for inaccuracies in matrices **A** and **B** and the lack of the initial error. The estimation error or observation error is the difference between the measured output and the estimated output. The initial error is the difference between the initial state and the initial estimated state. Thus, we define the mathematical model of the observer to be

Full order State observer

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \mathbf{A}\tilde{\mathbf{x}} + \mathbf{B}u + \mathbf{K}_e(y - \mathbf{C}\tilde{\mathbf{x}}) \\ &= (\mathbf{A} - \mathbf{K}_e\mathbf{C})\tilde{\mathbf{x}} + \mathbf{B}u + \mathbf{K}_ey\end{aligned}\tag{10-57}$$

where $\tilde{\mathbf{x}}$ is the estimated state and $\mathbf{C}\tilde{\mathbf{x}}$ is the estimated output. The inputs to the observer are the output y and the control input u . Matrix $\mathbf{K}_e$, which is called the observer gain matrix, is a weighting matrix to the correction term involving the difference between the measured output y and the estimated output $\mathbf{C}\tilde{\mathbf{x}}$. This term continuously corrects the model output and improves the performance of the observer. Figure 10-11 shows the block diagram of the system and the full-order state observer.

Full order State observer

Full-Order State Observer. The order of the state observer that will be discussed here is the same as that of the plant. Assume that the plant is defined by Equations (10-55) and (10-56) and the observer model is defined by Equation (10-57).

To obtain the observer error equation, let us subtract Equation (10-57) from Equation (10-55):

$$\begin{aligned}\dot{\mathbf{x}} - \dot{\tilde{\mathbf{x}}} &= \mathbf{A}\mathbf{x} - \mathbf{A}\tilde{\mathbf{x}} - \mathbf{K}_e(\mathbf{C}\mathbf{x} - \mathbf{C}\tilde{\mathbf{x}}) \\ &= (\mathbf{A} - \mathbf{K}_e\mathbf{C})(\mathbf{x} - \tilde{\mathbf{x}})\end{aligned}\quad (10-58)$$

Define the difference between $\mathbf{x}$ and $\tilde{\mathbf{x}}$ as the error vector $\mathbf{e}$, or

$$\mathbf{e} = \mathbf{x} - \tilde{\mathbf{x}}$$

Then Equation (10-58) becomes

$$\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{K}_e\mathbf{C})\mathbf{e}\quad (10-59)$$

Full order State observer

From Equation (10–59), we see that the dynamic behavior of the error vector is determined by the eigenvalues of matrix $\mathbf{A} - \mathbf{K}_e \mathbf{C}$. If matrix $\mathbf{A} - \mathbf{K}_e \mathbf{C}$ is a stable matrix, the error vector will converge to zero for any initial error vector $\mathbf{e}(0)$. That is, $\tilde{\mathbf{x}}(t)$ will converge to $\mathbf{x}(t)$ regardless of the values of $\mathbf{x}(0)$ and $\tilde{\mathbf{x}}(0)$. If the eigenvalues of matrix $\mathbf{A} - \mathbf{K}_e \mathbf{C}$ are chosen in such a way that the dynamic behavior of the error vector is asymptotically stable and is adequately fast, then any error vector will tend to zero (the origin) with an adequate speed.

If the plant is completely observable, then it can be proved that it is possible to choose matrix $\mathbf{K}_e$ such that $\mathbf{A} - \mathbf{K}_e \mathbf{C}$ has arbitrarily desired eigenvalues. That is, the observer gain matrix $\mathbf{K}_e$ can be determined to yield the desired matrix $\mathbf{A} - \mathbf{K}_e \mathbf{C}$. We shall discuss this matter in what follows.

Design of State observers

Transformation Approach to Obtain State Observer Gain Matrix $\mathbf{K}_e$. By following the same approach as we used in deriving the equation for the state feedback gain matrix $\mathbf{K}$, we can obtain the following equation:

$$\mathbf{K}_e = \mathbf{Q} \begin{bmatrix} \alpha_n - a_n \\ \alpha_{n-1} - a_{n-1} \\ \vdots \\ \alpha_1 - a_1 \end{bmatrix} = (\mathbf{W}\mathbf{N}^*)^{-1} \begin{bmatrix} \alpha_n - a_n \\ \alpha_{n-1} - a_{n-1} \\ \vdots \\ \alpha_1 - a_1 \end{bmatrix} \quad (10-61)$$

where $\mathbf{K}_e$ is an $n \times 1$ matrix,

$$\mathbf{Q} = (\mathbf{W}\mathbf{N}^*)^{-1}$$

and

$$\mathbf{N} = [\mathbf{C}^* \mid \mathbf{A}^*\mathbf{C}^* \mid \cdots \mid (\mathbf{A}^*)^{n-1}\mathbf{C}^*]$$

$$\mathbf{W} = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & a_{n-3} & \cdots & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_1 & 1 & \cdots & 0 & 0 \end{bmatrix}$$

Design of State observers

Direct-Substitution Approach to Obtain State Observer Gain Matrix $\mathbf{K}_e$.

Similar to the case of pole placement, if the system is of low order, then direct substitution of matrix $\mathbf{K}_e$ into the desired characteristic polynomial may be simpler. For example, if $\mathbf{x}$ is a 3-vector, then write the observer gain matrix $\mathbf{K}_e$ as

$$\mathbf{K}_e = \begin{bmatrix} k_{e1} \\ k_{e2} \\ k_{e3} \end{bmatrix}$$

Substitute this $\mathbf{K}_e$ matrix into the desired characteristic polynomial:

$$|s\mathbf{I} - (\mathbf{A} - \mathbf{K}_e\mathbf{C})| = (s - \mu_1)(s - \mu_2)(s - \mu_3)$$

By equating the coefficients of the like powers of s on both sides of this last equation, we can determine the values of k_{e1} , k_{e2} , and k_{e3} . This approach is convenient if $n = 1, 2$, or 3 , where n is the dimension of the state vector $\mathbf{x}$. (Although this approach can be used when $n = 4, 5, 6, \dots$, the computations involved may become very tedious.)

Another approach to the determination of the state observer gain matrix $\mathbf{K}_e$ is to use Ackermann's formula. This approach is presented in the following.

Design of State observers

Ackermann's Formula. Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (10-62)$$

$$y = \mathbf{C}\mathbf{x} \quad (10-63)$$

In Section 10-2 we derived Ackermann's formula for pole placement for the system defined by Equation (10-62). The result was given by Equation (10-18), rewritten thus:

$$\mathbf{K} = [0 \quad 0 \cdots 0 \quad 1][\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}]^{-1}\phi(\mathbf{A})$$

For the dual of the system defined by Equations (10-62) and (10-63),

$$\dot{\mathbf{z}} = \mathbf{A}^*\mathbf{z} + \mathbf{C}^*v$$

$$n = \mathbf{B}^*\mathbf{z}$$

the preceding Ackermann's formula for pole placement is modified to

$$\mathbf{K} = [0 \quad 0 \cdots 0 \quad 1][\mathbf{C}^* \quad \mathbf{A}^*\mathbf{C}^* \quad \cdots \quad (\mathbf{A}^*)^{n-1}\mathbf{C}^*]^{-1}\phi(\mathbf{A}^*) \quad (10-64)$$

Problem

Consider the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 20.6 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

We use the observed state feedback such that

$$u = -\mathbf{K}\tilde{\mathbf{x}}$$

Problem

Design a full-order state observer, assuming that the system configuration is identical to that shown in Figure 10–11. Assume that the desired eigenvalues of the observer matrix are

$$\mu_1 = -10, \quad \mu_2 = -10$$

The design of the state observer reduces to the determination of an appropriate observer gain matrix $\mathbf{K}_e$.

Let us examine the observability matrix. The rank of

$$[\mathbf{C}^* \mid \mathbf{A}^* \mathbf{C}^*] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is 2. Hence, the system is completely observable and the determination of the desired observer gain matrix is possible. We shall solve this problem by three methods.

Method 1: We shall determine the observer gain matrix by use of Equation (10–61). The given system is already in the observable canonical form. Hence, the transformation matrix $\mathbf{Q} = (\mathbf{W}\mathbf{N}^*)^{-1}$ is $\mathbf{I}$. Since the characteristic equation of the given system is

$$|s\mathbf{I} - \mathbf{A}| = \begin{vmatrix} s & -20.6 \\ -1 & s \end{vmatrix} = s^2 - 20.6 = s^2 + a_1s + a_2 = 0$$

Problem

we have

$$a_1 = 0, \quad a_2 = -20.6$$

The desired characteristic equation is

$$(s + 10)^2 = s^2 + 20s + 100 = s^2 + \alpha_1 s + \alpha_2 = 0$$

Hence,

$$\alpha_1 = 20, \quad \alpha_2 = 100$$

Then the observer gain matrix $\mathbf{K}_e$ can be obtained from Equation (10-61) as follows:

$$\mathbf{K}_e = (\mathbf{W}\mathbf{N}^*)^{-1} \begin{bmatrix} \alpha_2 - a_2 \\ \alpha_1 - a_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 100 + 20.6 \\ 20 - 0 \end{bmatrix} = \begin{bmatrix} 120.6 \\ 20 \end{bmatrix}$$

Problem

Method 2: Referring to Equation (10–59):

$$\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{K}_e \mathbf{C})\mathbf{e}$$

the characteristic equation for the observer becomes

$$|s\mathbf{I} - \mathbf{A} + \mathbf{K}_e \mathbf{C}| = 0$$

Define

$$\mathbf{K}_e = \begin{bmatrix} k_{e1} \\ k_{e2} \end{bmatrix}$$

Then the characteristic equation becomes

$$\begin{aligned} & \left| \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 20.6 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} k_{e1} \\ k_{e2} \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \right| = \begin{vmatrix} s & -20.6 + k_{e1} \\ -1 & s + k_{e2} \end{vmatrix} \\ & = s^2 + k_{e2}s - 20.6 + k_{e1} = 0 \end{aligned}$$

Problem

Since the desired characteristic equation is

$$s^2 + 20s + 100 = 0$$

by comparing Equation (10–66) with this last equation, we obtain

$$k_{e1} = 120.6, \quad k_{e2} = 20$$

or

$$\mathbf{K}_e = \begin{bmatrix} 120.6 \\ 20 \end{bmatrix}$$

Problem

Method 3: We shall use Ackermann's formula given by Equation (10-65):

$$\mathbf{K}_e = \phi(\mathbf{A}) \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

where

$$\phi(s) = (s - \mu_1)(s - \mu_2) = s^2 + 20s + 100$$

Thus,

$$\phi(\mathbf{A}) = \mathbf{A}^2 + 20\mathbf{A} + 100\mathbf{I}$$

and

$$\begin{aligned} \mathbf{K}_e &= (\mathbf{A}^2 + 20\mathbf{A} + 100\mathbf{I}) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 120.6 & 412 \\ 20 & 120.6 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 120.6 \\ 20 \end{bmatrix} \end{aligned}$$

Problem

As a matter of course, we get the same $\mathbf{K}_e$ regardless of the method employed.

The equation for the full-order state observer is given by Equation (10-57),

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{K}_e \mathbf{C}) \tilde{\mathbf{x}} + \mathbf{B}u + \mathbf{K}_e y$$

or

$$\begin{bmatrix} \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \end{bmatrix} = \begin{bmatrix} 0 & -100 \\ 1 & -20 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 120.6 \\ 20 \end{bmatrix} y$$

Finally, it is noted that, similar to the case of pole placement, if the system order n is 4 or higher, methods 1 and 3 are preferred, because all matrix computations can be carried out by a computer, while method 2 always requires hand computation of the characteristic equation involving unknown parameters $k_{e1}, k_{e2}, \dots, k_{en}$.

THANK YOU